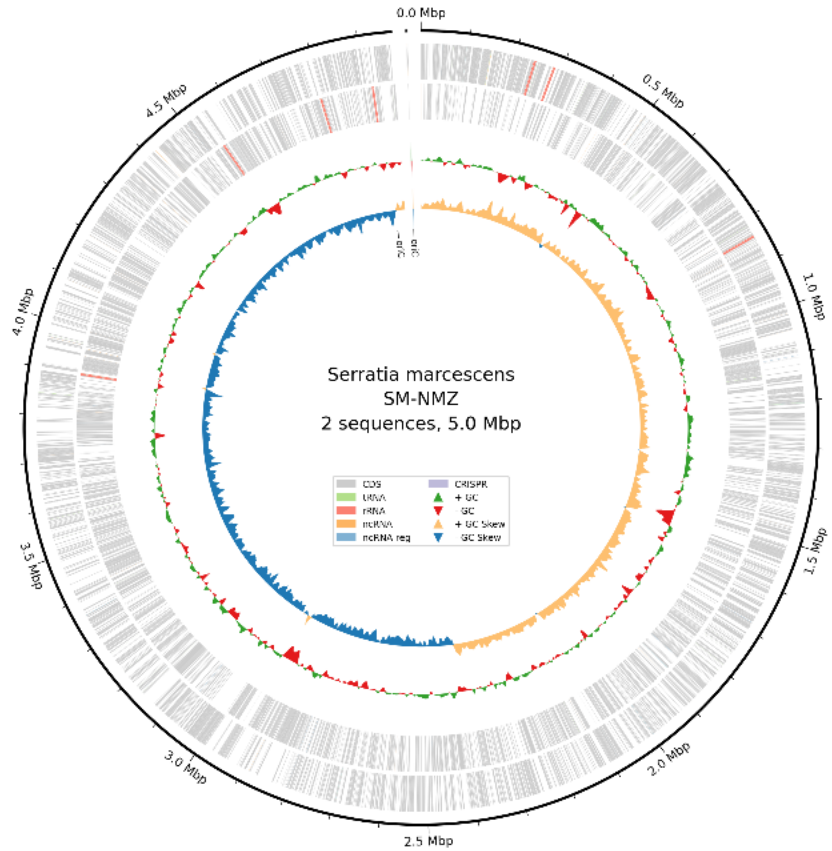
NMZ Serratia genomics

An explainer for the research group

SM-NMZ Slaughterhouse effluent
C14-NMZ Calotropis rhizosphere



6 November 2026

DNA, gene predictions and activity

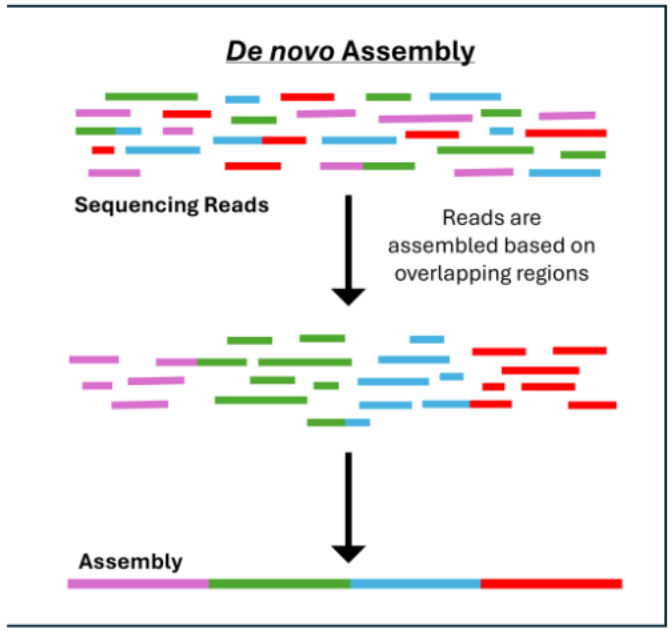
Stage	What it supplies
Sequencing	DNA fragments called reads
Assembly	Reconstructed chromosome and other replicons
Annotation	Predicted genes and possible functions
Experiment	An observed effect under stated conditions

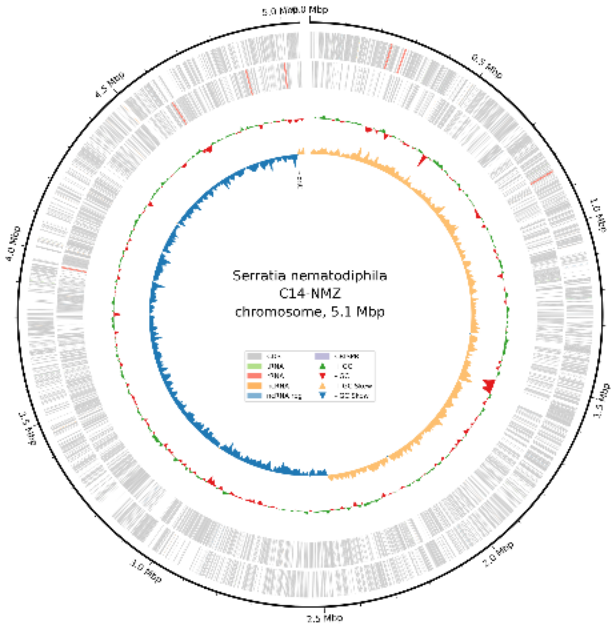
Gene predictions give us candidates to compare with lab assays.

Reads, assembly and genome annotation

Basecalled reads → Draft assembly → Annotated genome

```
@SRR014849.1 EIXKN4201CFU84 length=93
GGGGGGGGGGGGGGGGCTTTTTTTGTTTGGAACCGAAAGG
GTTTTGAATTTCAAACCCTTTTCGGTTTCCAACCTTCCAA
AGCAATGCCAATA
+SRR014849.1 EIXKN4201CFU84 length=93
3+&$#""""""""""""7F@71,'";C?,B;?6B;:EA1EA
1EA5'9B:?:#9EA0D@2EA5':>5?:%A;A8A;?9B;D@
/=<?7=9<2A8==
```





DNA bases + quality

Overlaps → contigs

Features + predictions

EMBL-EBI examples (left, centre) · C14 Bakta map (right): GTDB S. nematodiphila (slide 5)

What each kind of evidence answers

Evidence	Question it addresses
Related gene or protein	Is there a plausible candidate?
Several linked genes	Does the proposed machinery fit together?
Chemical analysis	Which molecular assignment has support?
Activity assay	What effect occurred in this experiment?
Controlled perturbation	Does this factor contribute to that effect?

Genome resources and identity

Feature	SM-NMZ	C14-NMZ
Chromosome (closed circular)	4.971 Mb	5.051 Mb
Additional replicon	3.581-kb circular plasmid	None recovered
Bakta CDSs	4,566	4,626
GTDB r232 label	S. marcescens	S. nematodiphila
TYGS species / subspecies cluster	6 / 11	6 / 10

Closed circular chromosomes avoid contig breaks across our 35–44 kb BGC neighborhoods.

TYGS groups both in S. marcescens cluster 6; Bakta and clinker plots retain C14-NMZ's GTDB label (S. nematodiphila).

ANI and digital DNA–DNA hybridization

Measure	What is compared	How it is interpreted
ANI + aligned fraction	DNA sequence identity in aligned regions	Identity together with alignment coverage
dDDH	Sequence-based estimate of DNA–DNA relatedness	Estimate and confidence interval

ANI = average nucleotide identity; AF = aligned fraction

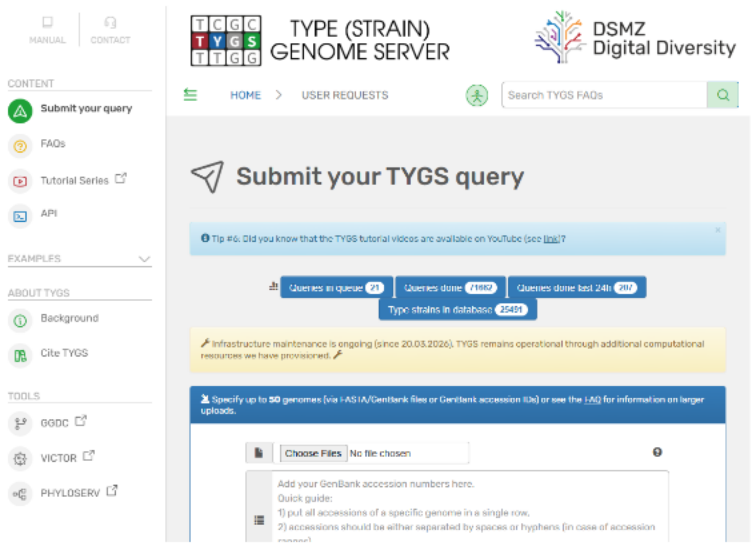
Broad species guides: ANI ≈95–96%; dDDH ≈70%

GTDB applies a representative-specific ANI radius.

NMZ pair: dDDH(d4) 74.2% (71.2–77.0%)

GTDB-Tk and TYGS workflows

TYGS · web service



Genome FASTA → type-strain comparisons
GBDP → dDDH + genome phylogram

GTDB-Tk · local pipeline

GTDB = Genome Taxonomy Database

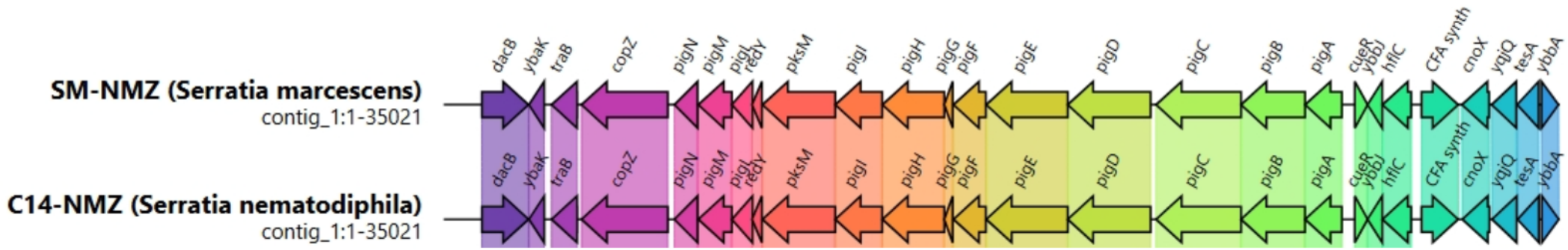
Genome FASTA + GTDB references
↓
ANI + aligned fraction
↓
Representative-specific species boundary
↓
Classification summary

NMZ queries were classified by ANI screening.

C14: GTDB S. nematodiphila; TYGS S. marcescens

Genes, neighborhoods, clusters and operons

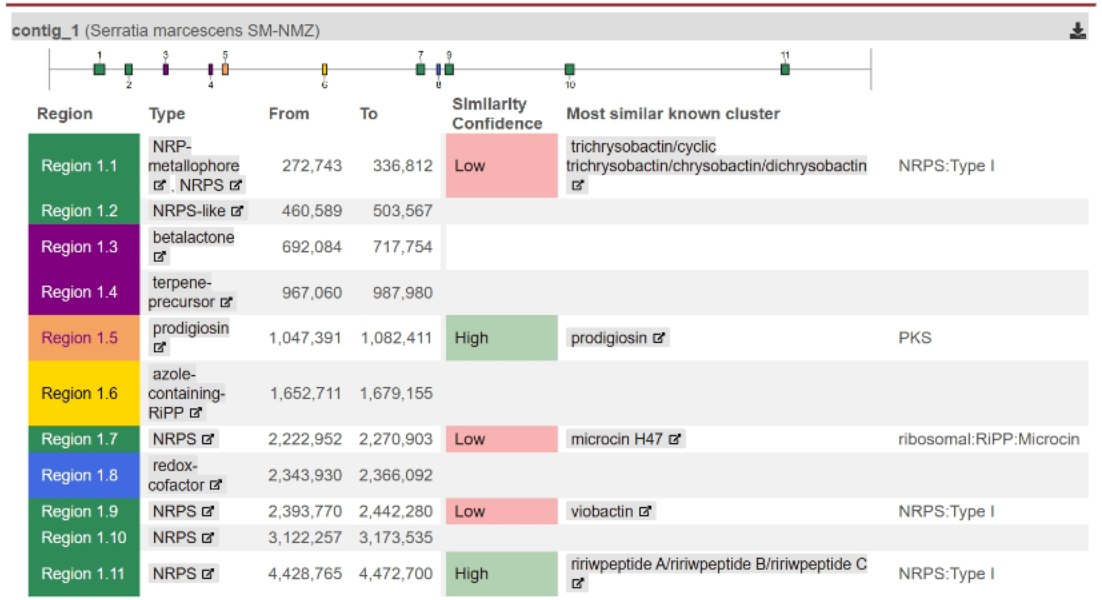
Term	Meaning
Gene	DNA sequence encoding a product
Neighborhood	Nearby genes: physical context
Biosynthetic cluster	Nearby genes implicated in making a molecule
Operon	Genes transcribed together as a unit



pig: a multi-gene example. SwrW: a multifunctional enzyme encoded by one gene.

Genome-wide BGC mining with antiSMASH

SM-NMZ · 11 BGC regions (4 NRPS + 1 NRPS-like)



Region	Type	From	To	Similarity Confidence	Most similar known cluster	
Region 1.1	NRP-metallophore, NRPS	272,743	336,812	Low	trichrysobactin/cyclic trichrysobactin/chrysobactin/dichrysobactin	NRPS:Type I
Region 1.2	NRPS-like	460,589	503,567			
Region 1.3	betalactone	692,084	717,754			
Region 1.4	terpene-precursor	967,060	987,980			
Region 1.5	prodigiosin	1,047,391	1,082,411	High	prodigiosin	PKS
Region 1.6	azole-containing-RiPP	1,652,711	1,679,155			
Region 1.7	NRPS	2,222,952	2,270,903	Low	microcin H47	ribosomal:RiPP:Microcin
Region 1.8	redox-cofactor	2,343,930	2,366,092			
Region 1.9	NRPS	2,393,770	2,442,280	Low	viobactin	NRPS:Type I
Region 1.10	NRPS	3,122,257	3,173,535			
Region 1.11	NRPS	4,428,765	4,472,700	High	ririwpeptide A/ririwpeptide B/ririwpeptide C	NRPS:Type I

C14-NMZ · 11 BGC regions (3 NRPS + RiPP + opine)



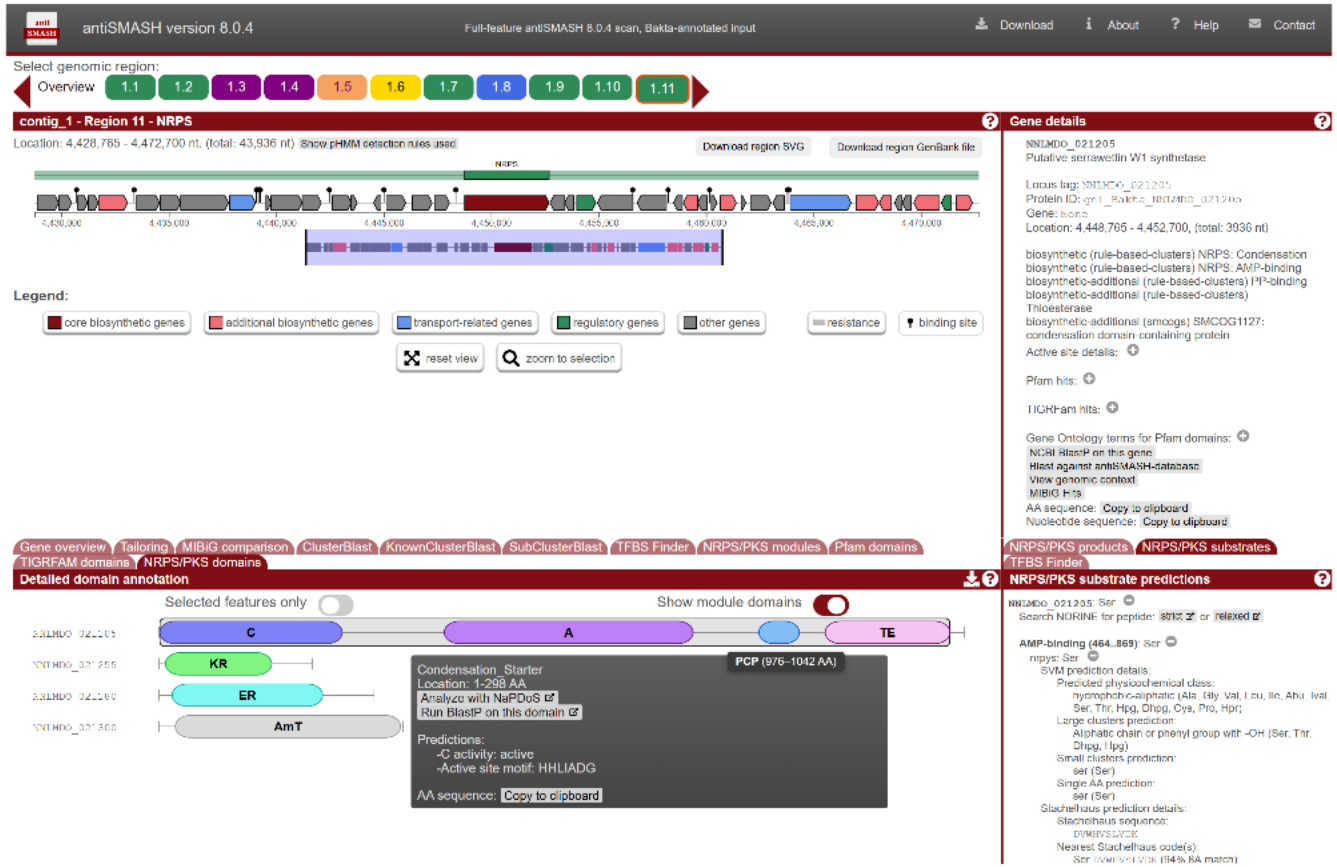
Region	Type	From	To	Similarity Confidence	Most similar known cluster	
Region 1	NRP-metallophore, NRPS	268,929	332,990	Low	trichrysobactin/cyclic trichrysobactin/chrysobactin/dichrysobactin	NRPS:Type I
Region 2	betalactone	690,276	715,946			
Region 3	ranthipeptide	883,725	905,132			
Region 4	terpene-precursor	977,361	998,281			
Region 5	prodigiosin	1,057,699	1,092,719	High	prodigiosin	PKS
Region 6	azole-containing-RiPP	1,698,594	1,725,038			
Region 7	opine-like-metallophore	2,094,781	2,116,878	High	yersinopine	other:other
Region 8	redox-cofactor	2,403,996	2,426,158			
Region 9	NRPS	2,442,839	2,491,349	Low	viobactin	NRPS:Type I
Region 10	NRPS	3,163,455	3,209,241			
Region 11	NRPS	4,500,257	4,544,195			

Shared: prodigiosin (Reg 5), SwrW (Reg 11), PQQ (Reg 8), siderophores (Reg 1 & 9), RiPP/terpene/β-lactone.

Differences: SM-only Reg 2 & 7 (NRPS) · C14-only Reg 3 (ranthipeptide) & 7 (opine) · divergent Reg 10 NRPSs.

Three similarity lenses: 1) Overview badge (% ref genes hit) · 2) MIBiG score (composite geometric mean) · 3) ClusterBlast (full synteny).

Orienting in antiSMASH: Region 11 (swrW) full view



1) Top: 43.9-kb rule-defined window (20 kb flank + 3,936 nt core swrW + 20 kb flank; not an operon boundary).

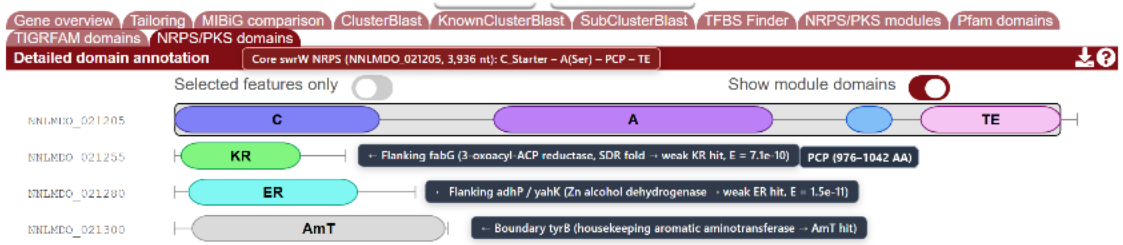
2) Bottom-left (“Show module domains” ON): C_Starter (1–298 AA, HHLIADG) – A – PCP (976–1042 AA) – TE module domains.

3) Right panel: Gene details & Ser substrate prediction (DVWHVSLVDK 100% Stachelhaus code match to L-Serine).

* Overlays added for clarity; all values verified against antiSMASH JSON/TSV outputs.

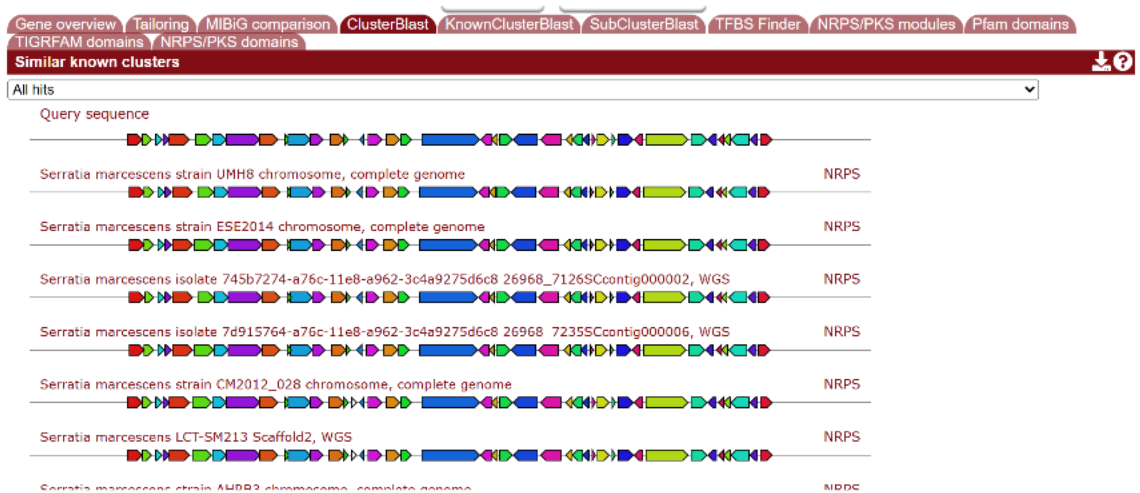
Region 11: core SwrW vs. flanking window hits

NRPS/PKS domains: core SwrW vs. 3 flanking enzymes



Why 4 rows appear on the left: antiSMASH scans the 20-kb flanking window, flagging housekeeping SDR/dehydrogenases (fabG, adhP) and an aminotransferase (tyrB).

ClusterBlast: 100% identity & synteny in S. marcescens



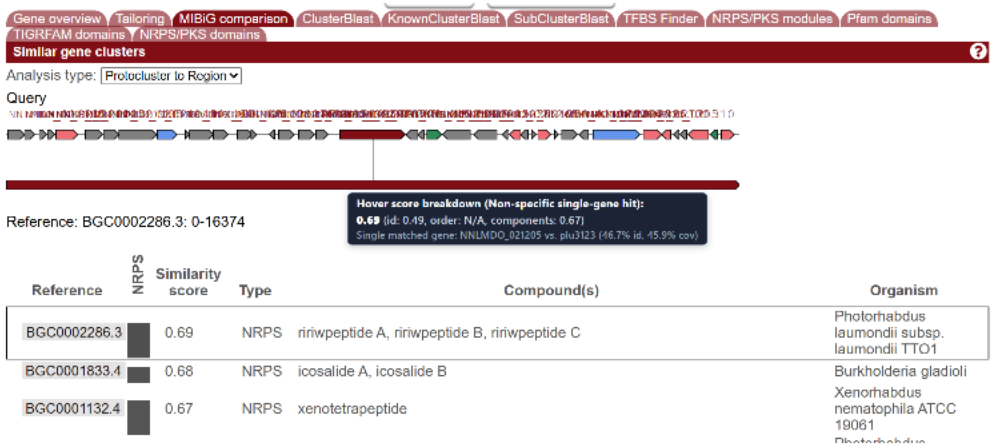
Only NNLMDO_021205 (SM) and OJMJKD_021655 (C14) encode the C_Starter–A(Ser)–PCP–TE SwrW synthetase.

ClusterBlast confirms 100% coverage and neighborhood collinearity with reference S. marcescens chromosomes.

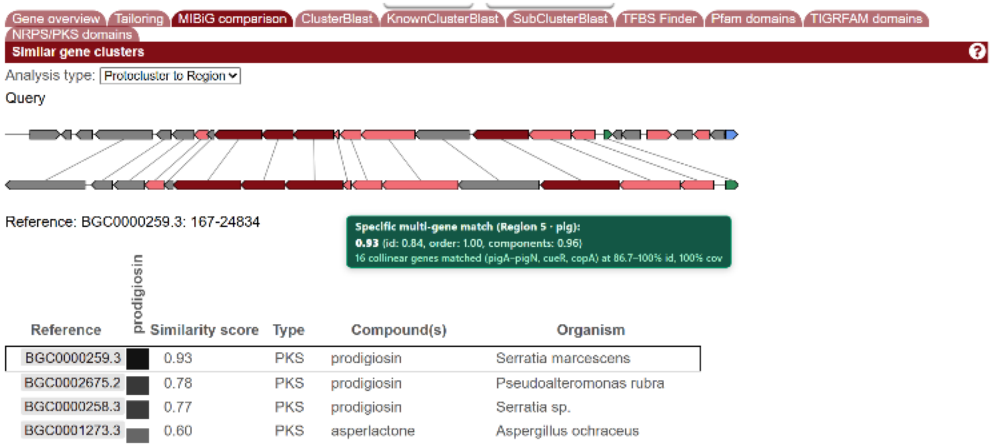
* Overlays added for clarity; domain E-values & ClusterBlast verified against antiSMASH outputs.

Region 11 (0.69) vs. Region 5 (0.93) MIBiG scores

Region 11 (swrW): 0.69 score from 1 generic NRPS hit (46.7% id)



Region 5 (pig): 0.93 score from 16 collinear genes (87–100% id)



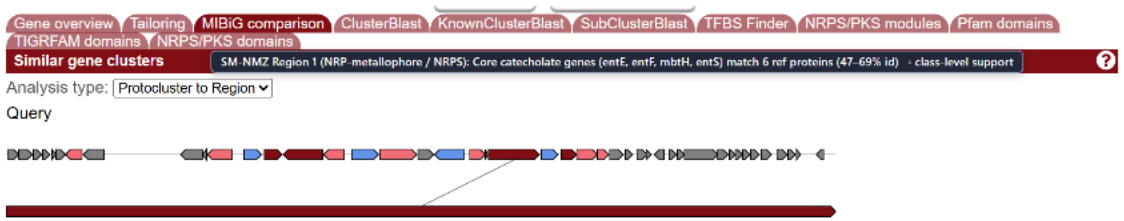
Similarity Lens	Region 11 (swrW vs. ririwpeptide)	Region 5 (pig vs. prodigiosin)	What the Number Actually Measures
1) Overview badge	High (green badge)	High (green badge)	Hits ÷ ref genes: 1/1 (100%) vs. 16/16 (100%); 1 gene inflates badge
2) MIBiG score	0.69 (id 0.49, order N/A, comp 0.67)	0.93 (id 0.84, order 1.00, comp 0.96)	Geometric mean: single-gene ref pads order to 1.0 (true 2-term is 0.57)
3) Protein alignment	46.7% id / 45.9% cov (1 generic NRPS)	86.7–100% id / 100% cov (16 genes)	Generic C–A–PCP–TE homology vs. complete collinear pathway synteny

* Overlays added for clarity; single-gene reference arithmetic verified against antiSMASH v8 source code.

Takeaway: Always inspect gene count and alignment depth—a single-gene hit inflates both overview and MIBiG scores.

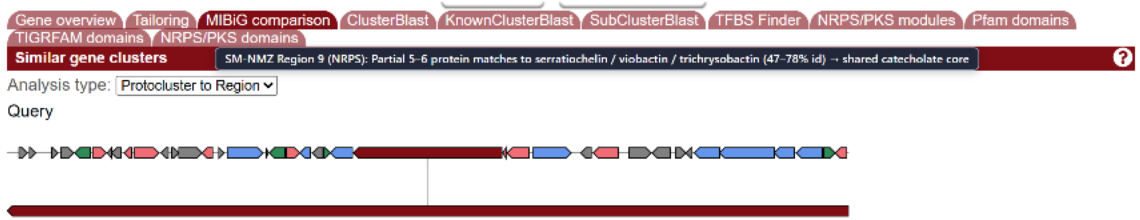
Catecholate siderophore potential: Regions 1 and 9

Region 1 (NRP-metallophore): entE, entF, mbtH, entS



Reference	NRP-metallophore	NRPS	Similarity score	Type	Compound(s)	Organism
BGC0002075.3	■		0.86	NRPS	pyreudione A, pyreudione B, pyreudione C, pyreudione D, pyreudione E	Pseudomonas fluorescens
BGC0000343.5		■	0.74	NRPS	enterobactin	Pseudomonas sp. J465

Region 9 (NRPS): 2nd catecholate locus (5–6 gene hits)



Reference	NRPS	Similarity score	Type	Compound(s)	Organism
BGC0002075.3	■	0.73	NRPS	pyreudione A, pyreudione B, pyreudione C, pyreudione D, pyreudione E	Pseudomonas fluorescens
BGC0001132.4	■	0.67	NRPS	xenotetrapeptide	Xenorhabdus nematophila ATCC 19061
BGC0001833.4	■	0.67	NRPS	icosalide A, icosalide B	Burkholderia gladioli
BGC0001128.4	■	0.66	NRPS	gamexpeptide C	Photorhabdus laumondii subsp. laumondii TTO1
BGC0002286.3	■	0.66	NRPS	ririwpeptide A, ririwpeptide B, ririwpeptide C	Photorhabdus laumondii subsp. laumondii TTO1

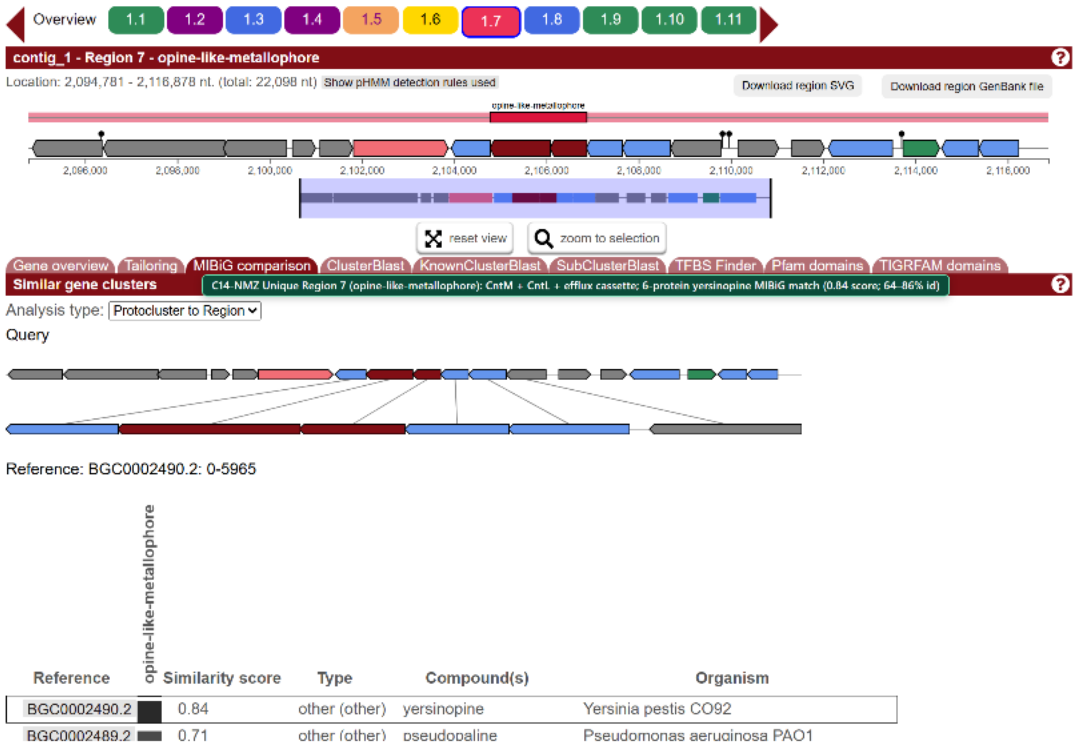
Both chromosomes carry two conserved catecholate siderophore loci (entE, entF/entF2, mbtH, entS).

Partial 5–6 protein MIBiG hits (47–78% identity) support catecholate class potential without naming a product.

* Overlays added for clarity; all values verified against antiSMASH JSON/TSV outputs.

C14 Region 7 (opine) vs. SM Regions 2 & 7 (NRPS)

C14-NMZ Region 7: opine-like metallophore (CntM + CntL)



SM-NMZ Region 2 (NRPS-like) & Region 7 (2-module NRPS)



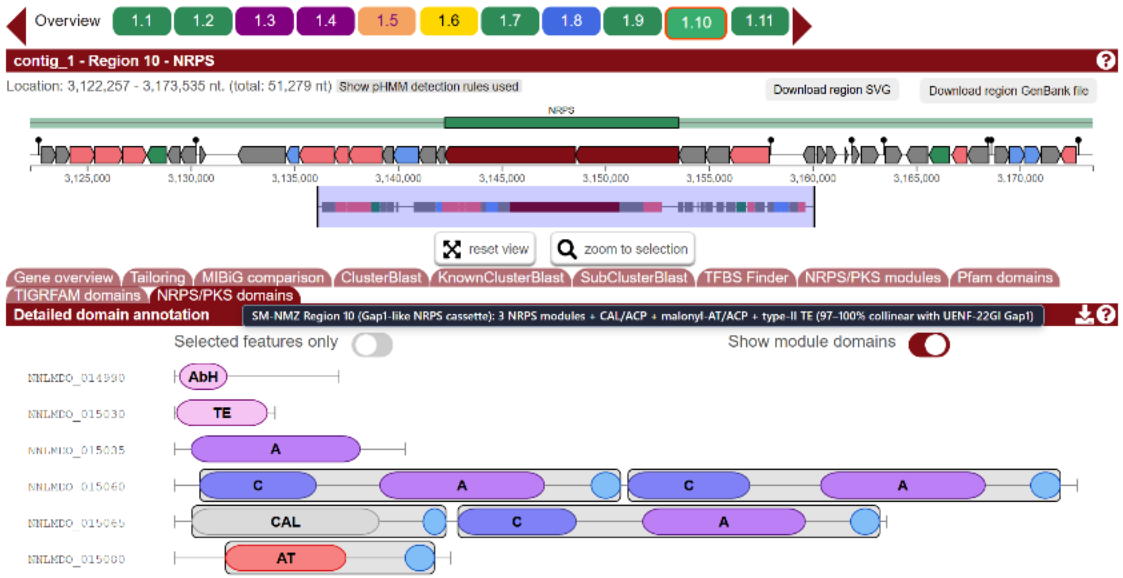
Symmetric divergence: SM-NMZ has 4 NRPS + 1 NRPS-like regions; C14-NMZ has 3 NRPS + opine + ranthipeptide.

All three isolate-specific loci stay as uncharacterized genomic candidates, without speculative compound names.

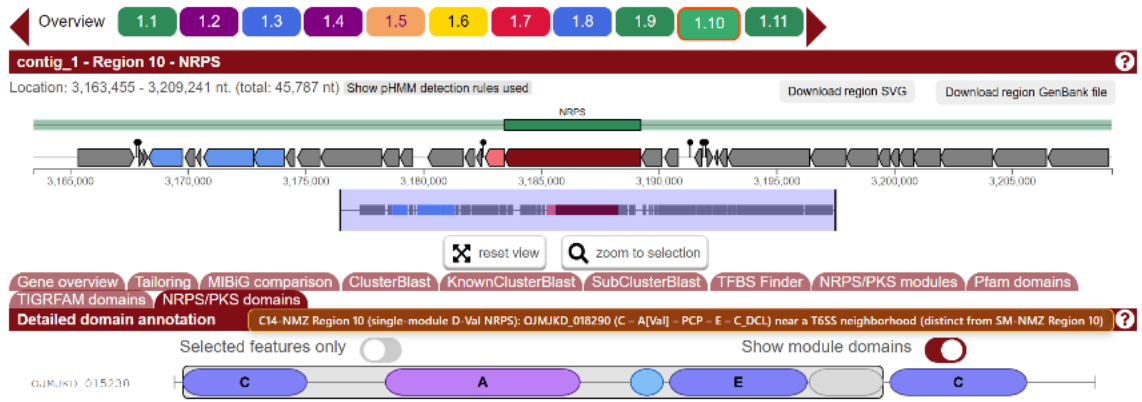
* Overlays added for clarity; all values verified against antiSMASH JSON/TSV outputs.

Different NRPS machinery at Region 10

SM-NMZ Region 10: Gap1-like 3-module NRPS cassette



C14-NMZ Region 10: single-module D-Val NRPS



SM-NMZ matches the UENF-22GI Gap1 cassette (97–100% id); C14-NMZ is a 1-module D-Val NRPS.

Neither architecture matches a canonical stephensiolide BGC—a structural check on LC-HRMS library hits.

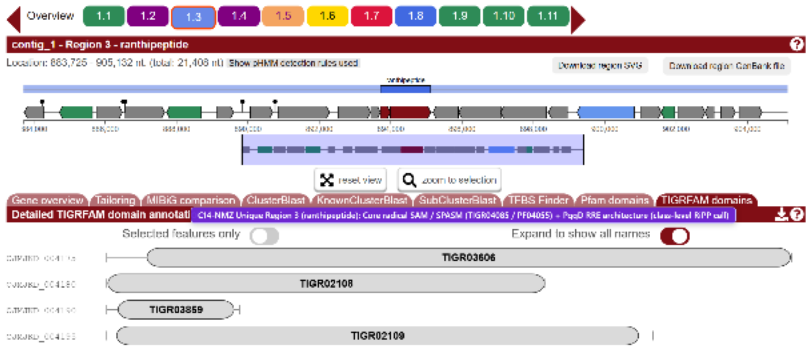
* Overlays added for clarity; all values verified against antiSMASH JSON/TSV outputs.

Class-level RiPP, PQQ, beta-lactone and terpene calls

Shared Region 8: conserved pqqA–pqqF PQQ operon



C14-NMZ Region 3 (ranthipeptide): rSAM/SPASM + RRE



Region (SM / C14)	antiSMASH Rule Call	Core Rule Signature	Manuscript Interpretation
Reg 8 / Reg 8	redox-cofactor	pqqA–pqqF operon (PqqD RRE + PqqE rSAM + PqqC)	Conserved PQQ redox-cofactor biosynthesis
— / Reg 3 (C14 only)	ranthipeptide	Radical SAM / SPASM (TIGR04085) + PqqD RRE	Class-level RiPP call unique to C14-NMZ
Reg 6 / Reg 6	azole-containing-RiPP	YcaO cyclodehydratase + PF04055 radical SAM	Conserved class-level RiPP call (no MIBiG hit)
Reg 3 / Reg 2 & Reg 4 / Reg 4	betalactone & terpene-precursor	HMGL-like + AMP-binding (β-lactone); ispG (terpene)	Conserved primary/secondary class rule calls

* Overlays added for clarity; all values verified against antiSMASH JSON/TSV outputs.

SwrW evidence in the NMZ study

Observation or test	What it supports
SwrW sequence, domains and context	A plausible biosynthetic candidate
Putative LC-HRMS assignments	Serrawettin W1 (= serratamolide A) and serratamolide C
Tension or E24 assay	Effect of the particular material tested
Controlled intervention / purified compound	A test of contribution to that effect

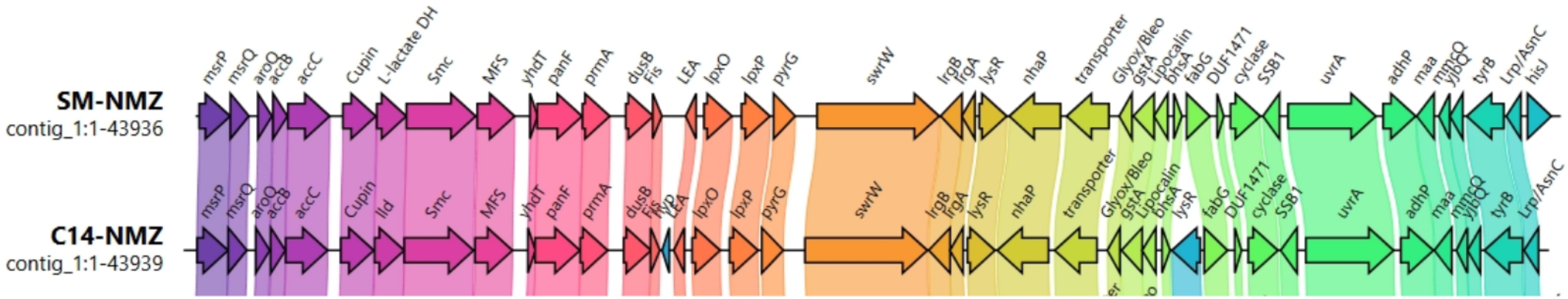
The three evidence layers are consistent; targeted assays test whether SwrW drives crude-oil activity.

Pair the chemistry with activity measured on the same test material.

SwrW and reported serratamolides

Genomic evidence

Both isolates carry a conserved swrW candidate.



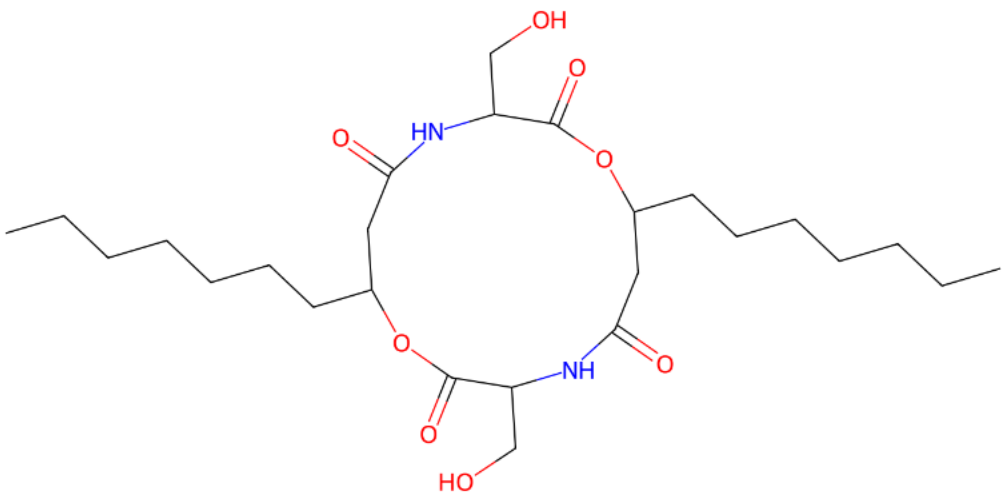
Putative LC-HRMS assignments · samples 1 and 2

Serrawettin W1 (= serratamolide A) and serratamolide C

The report does not map rows to isolates. Separately, pig fits the red/pink cultures.

Serratamolide A: the ring behind the LC-HRMS hit

Serratamolide A (= serrawettin W1): C26H46N2O8



Two 3-hydroxydecanoyl-Ser units close the ring through two amides and two esters.

What the mass spectrometer reports

Ion (calculated)	m/z
[M+H]+	515.3327
[M+Na]+	537.3146
[M+K]+ (workflow 1 only)	553.2886
+1 CH2 in an acyl chain	+14.0157

Chain-length congeners shift m/z in CH2 steps, so a library hit needs fragments or a standard.

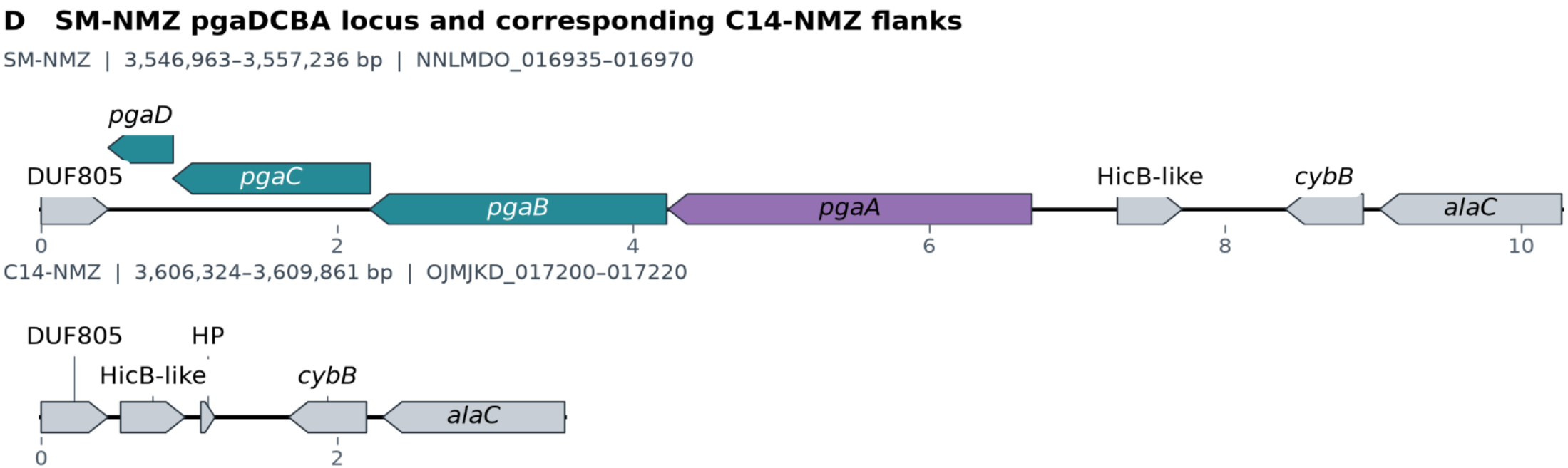
A mass fits a formula; fragments and standards test the structure.

The report lists serratamolides A-E plus library-only stephensiolides; B-E structures were not checked here.

Structure drawn with RDKit from SMILES; stereochemistry omitted.

Extracellular matrices and surface activity

Both isolates: capsule/EPS and cellulose candidates



Gene arrows. Scale in kb from the first displayed CDS.

SM-NMZ adds a contiguous pgaDCBA locus associated with PNAG.

Compare biofilm or emulsion stability if the isolates differ in those assays.

EPS, CPS and matrix polysaccharides

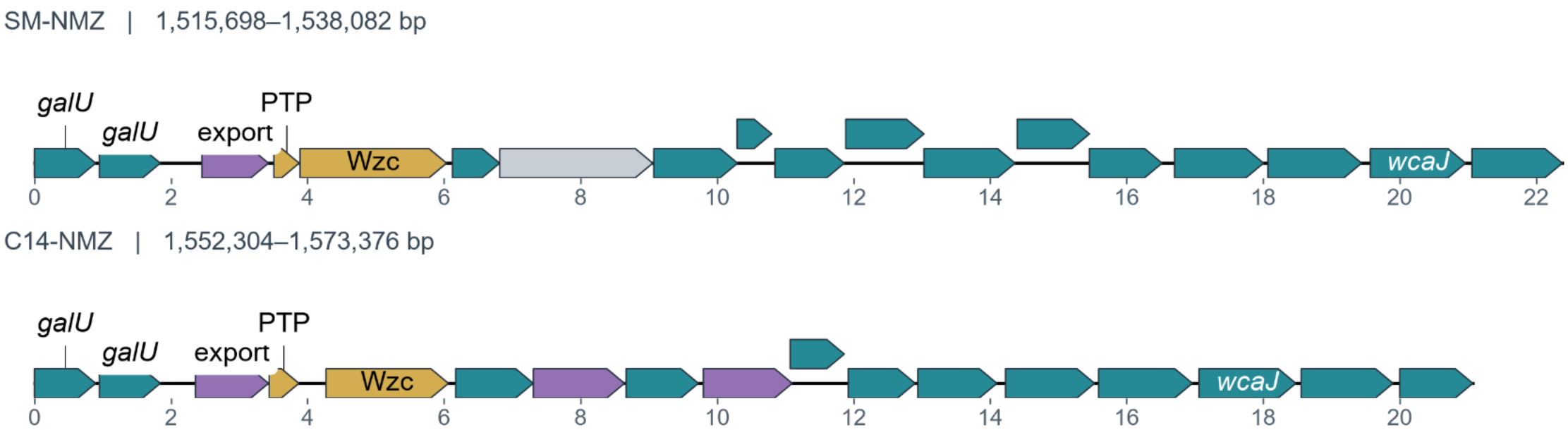
Polysaccharides are long chains of sugar units.

Term	What it means
EPS	Extracellular polymeric substances: the matrix
CPS	Capsular polysaccharide: associated with the cell
PNAG	β-(1→6)-linked N-acetylglucosamine polymer
Cellulose	β-(1→4)-linked glucose polymer

Small lipopeptides and matrix polymers can affect emulsions differently.

Record whether each assay used whole broth, supernatant or an extract.

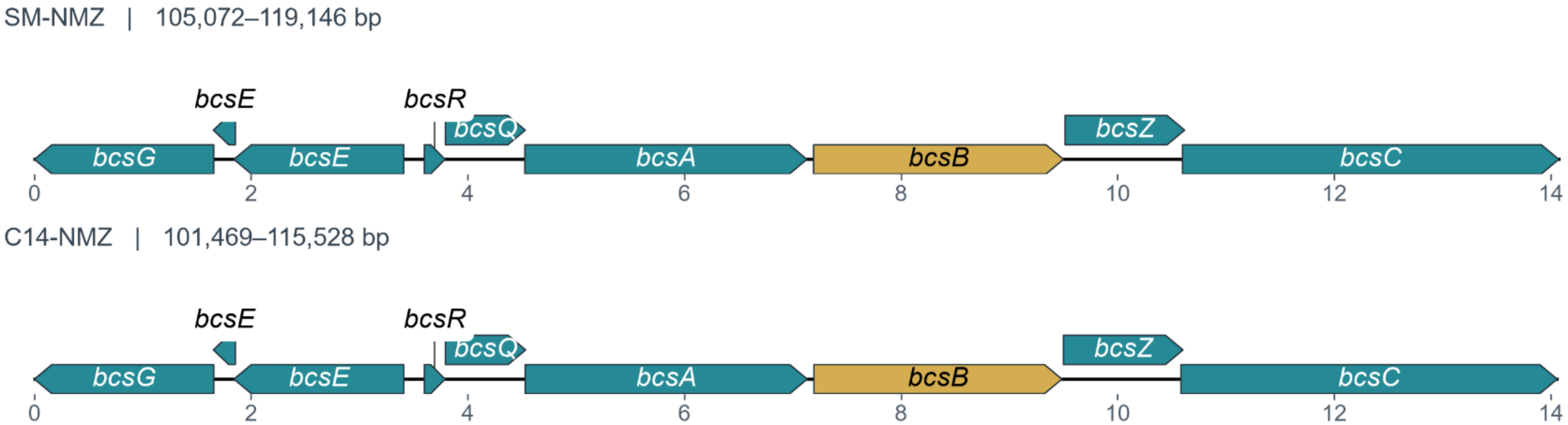
Capsule loci and the HADEG markers



Wzb-like PTP and Wzc are capsule-associated phosphatase and kinase candidates.

The HADEG “Emulsan” hits sit in capsule-associated neighborhoods.

Cellulose-associated loci in both isolates

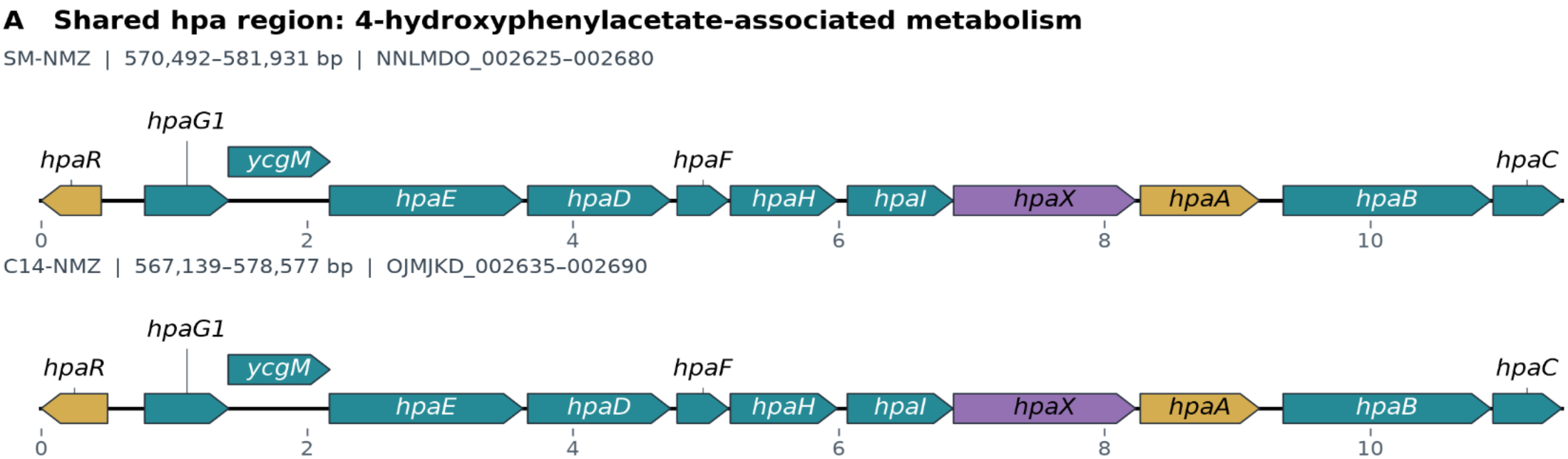


BcsA: synthesis · BcsB: partner · BcsC: export-associated

Scale: kb. Conserved gene order supports shared biosynthetic potential.

Aromatic metabolism depends on the substrate

Both isolates: organized hpa and paa loci



Gene arrows. Scale in kb from the first displayed CDS.

SM adds hydroxybenzoate/pca and gentisate candidates.

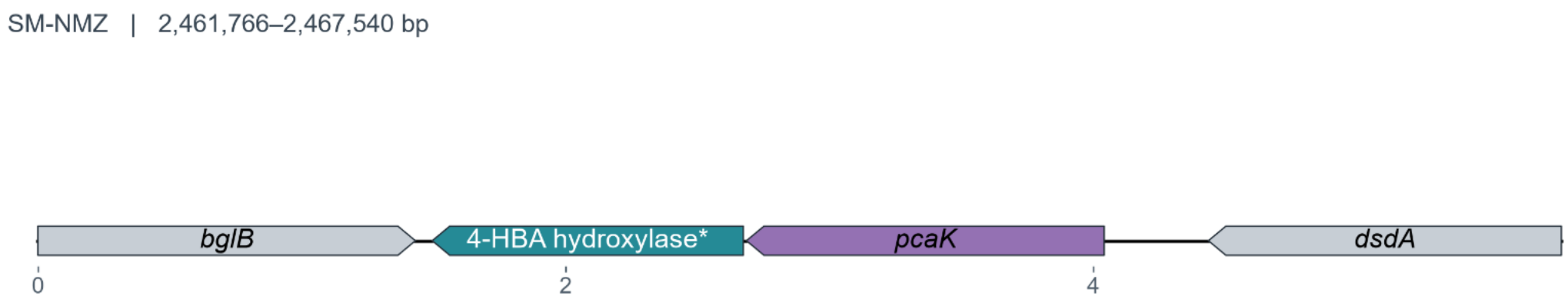
Match conversion assays to the oxygenated aromatic substrate and its route.

The HADEG Toluene_degradation hit

A characterized reference route · P. mendocina KR1

toluene → p-cresol → 4-HBA → protocatechuate

Initial activation and processing precede the PobA step.



SM NNLMDO_011765 and pcaK point to a downstream aromatic-acid step.

4-HBA = 4-hydroxybenzoate. *Label abbreviates the source product.

Aromatic hydrocarbons: no BTEX activation system

No complete upper-activation system was found; initiation of pristine petroleum solvents is not supported.

Hydrocarbon Class	Target Petroleum Substrates	Required Upper Activation Enzymes	Status in SM-NMZ & C14-NMZ	Biological / MEOR Reality
BTEX monoaromatics	Benzene, toluene, ethylbenzene, xylenes	Ring mono/dioxygenases (todABC1C2DE, tmoABCDEF, xylAM)	No complete system in either genome	Genomes do not support BTEX activation (activity untested)
Polycyclic aromatics (PAH)	Naphthalene, phenanthrene, biphenyl	Ring-hydroxylating dioxygenases (nah, ndo, bph, nid, phd)	No complete system in either genome	Genomes do not support PAH ring hydroxylation (activity untested)
Aliphatic petroleum alkanes	Medium/long-chain n-alkanes (C5–C16+)	Membrane alkane hydroxylases (alkB, CYP153, almA)	No convincing hydroxylase in either genome	Annotated alkB is DNA oxidative demethylase (K03919), not alkane hydroxylase

Petroleum hydrocarbon initiation needs upper oxygenases that were not found in either isolate.

The HADEG “Toluene_degradation” marker reflects downstream PobA homology, not toluene oxidation.

Aromatic scavenging: shared vs. SM-only cleavage

Central ring-cleavage pathways assimilate pre-oxygenated plant, amino acid, and waste aromatics.

Central Cleavage Pathway	Target Substrate & Natural Source	Key Enzyme Machinery	Isolate Distribution	Biological & Ecological Role
Homoprotocatechuate meta-cleavage	4-Hydroxyphenylacetate (plant lignin, tyrosine catabolism)	Complete 12-gene hpa cassette (hpaR, hpaX, hpaBC, hpaD–I)	CONSERVED in both (NNLMDO_002625–80; OJMJKD_002635–90)	Extradiol meta-cleavage yielding pyruvate and succinate semialdehyde
Phenylacetyl-CoA epoxidation	Phenylacetate (phenylalanine catabolism, organic waste)	Complete 14-gene paa block (paaK, paaABCDE, downstream ring processors)	CONSERVED in both (NNLMDO_014780–845; OJMJKD_015005–70)	CoA-linked ring epoxidation, then beta-oxidation-like steps
Protocatechuate ortho-cleavage	4-Hydroxybenzoate (plant phenolics, lignin fragments)	pobA–pcaK entry module + pcaIJFBDGH β-ketoadipate operon	SM-NMZ ONLY (NNLMDO_011765–70, 014970–5005)	Intradiol ortho-cleavage pathway expanding aromatic nutrition in SM-NMZ
Gentisate pathway	Gentisate / 2,5-dihydroxybenzoate (plant secondary metabolites)	maiA–fumaryl-intermediate hydrolase–gtdA cassette	SM-NMZ ONLY (NNLMDO_016045–65)	Alternative ring-cleavage branch feeding directly into fumarate & pyruvate

Shared core: hpa and paa cassettes are conserved; SM-NMZ adds pobA/pca and gentisate branches.

Reading: these loci fit oxygenated plant and amino-acid aromatics; crude-oil activity remains untested.

LadA and the alkane-oxidation hypothesis

Historically proposed role

Long-chain alkane → primary alcohol

LadA was reported as a soluble, flavin-dependent candidate for this first step.

Why search here?

To identify a plausible initiating enzyme alongside downstream carbon metabolism.

LadA evidence across the literature

Study	Evidence reported	Interpretation
Feng et al. 2007	Canonical LadA: in-vivo and in-vitro support	Initial long-chain oxidation proposed
Boonmak et al. 2014	B23 homologs: alkane depletion in a heterologous host	Cell-based evidence; not purified enzyme
Mireles et al. 2025	No direct hydroxylation detected for tested proteins	First-step assignment questioned

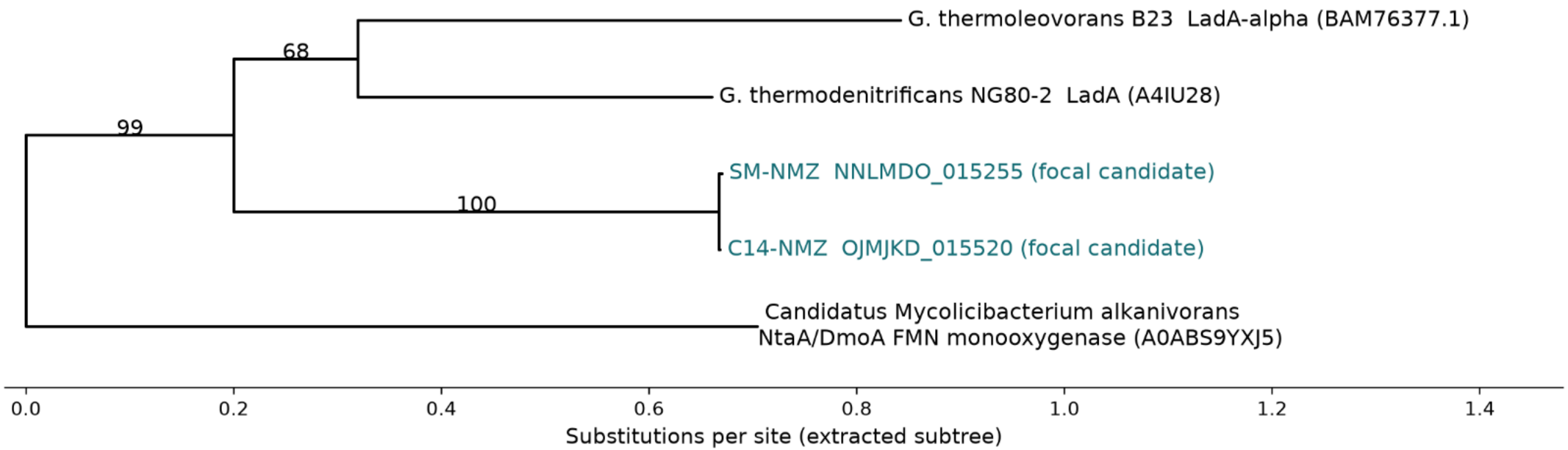
The literature gives us a candidate family and a substrate question to test.

Initial alkane oxidation

Simplified step	Genomic interpretation
Alkane to alcohol	Initiating system unresolved
Alcohol to aldehyde	Downstream processing candidates
Aldehyde to fatty acid	Downstream processing candidates
Fatty acid to cellular metabolism	Beta-oxidation and assimilation context

Both genomes encode β-oxidation. Alkane initiation remains unresolved.

The LadA-like comparison



Node labels show UFBoot support (%).

UFBoot support: 100% for the focal pair, 99% for the LadA-associated clade.

The conserved ssuD and metNPQ context adds a sulfur-related substrate hypothesis.

Pan-proteome: core, shared and unique groups



Group type	Groups	Share	How to read it
Core (all 14 genomes)	2,140	20.9%	Present in every genome, including the outgroup; typically basic cell functions
Shared (2-13 genomes)	4,096	40.0%	Accessory; example: the LadA-like K20938 group, found in 10 of 14 genomes
Genome-specific (1 genome)	4,016	39.2%	Unique genes, divergent genes below the cutoffs, or annotation differences
All groups	10,252	100%	66,241 proteins; standard ProteinOrtho run (no synteny filter)

Groups, not genes: 2,140 are universal, while 4,016 occur in a single proteome.

Sequence clusters show relatedness, not function; this panel cannot measure pan-genome openness.

Matching assays to genomic candidates

Endpoint, if measured	Relevant interpretation	Evidence to pair with it
Surface/interfacial tension	Amphiphile contribution	Material, interface, mN/m, control
E24 emulsification index	Amphiphile and/or matrix contribution	Oil, 24-h protocol, layer heights
Aromatic conversion	Route matching the exact substrate	Substrate, products and recovery controls
Pour point	Low-temperature oil-flow performance	Oil, dose, method and temperature

Pair each assay value with the material tested and its control.

Host-interaction genes

Reference-associated system	Biological role in characterized systems
ShlA–ShlB	Hemolysin machinery
Has	Heme acquisition
Type VI secretion system	Bacterial competition

VFDB sequence matches provide comparative context and do not quantify pathogenicity or establish environmental biosafety.
Deployment readiness requires dedicated antimicrobial susceptibility, virulence-expression and regulatory biosafety testing.

The genomics contribution: evidence scoreboard

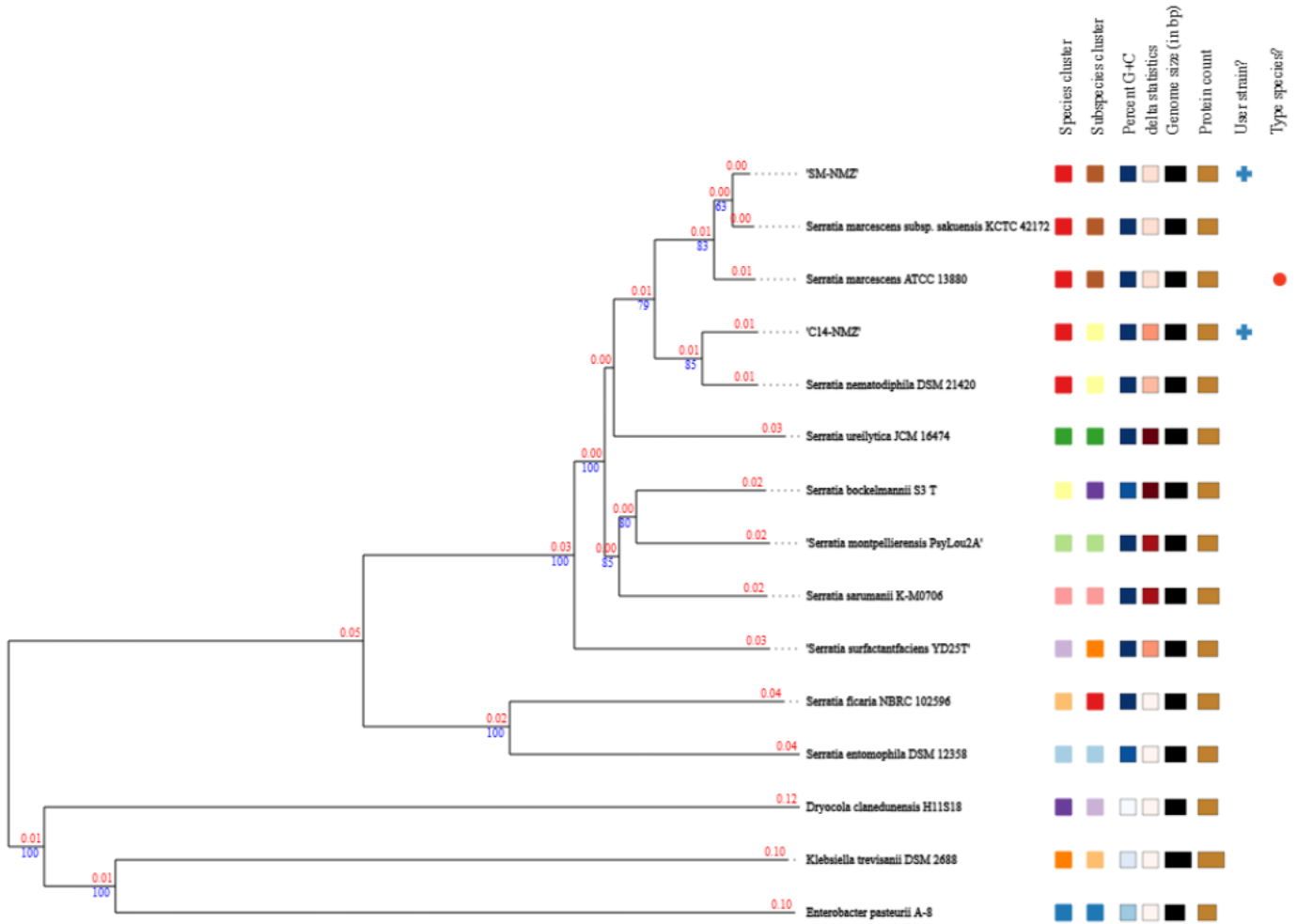
System / locus	Genomic candidate	Chemistry / phenotype in hand	Assay slot & decisive test
Biosurfactant (swrW)	Conserved 1-module NRPS (98.9% id)	LC-HRMS: serratamolide A (W1) & C	Tension, E24, pour point · ΔswrW + isolate LC-HRMS
Pigment (pigA–pigN)	Conserved 14-gene cluster (87–100% id)	Red/pink colony pigmentation	Phenotypic confirmation complete (pigment extract)
Biofilm / matrix	Shared CPS/EPS2/bcs + SM pgaDCBA	— (polymer chemistry unmeasured)	E24 stability · DspB (PNAG) & cellulase sensitivity
Oxygenated aromatics	Shared hpa/paa + SM pobA/pca/gtdA	—	Single-substrate growth (4-HBA, PCA, phenylacetate)
Alkane / monooxygenase	Fad β-oxidation + LadA-like (ssuD)	— (initiator unresolved)	Alkane GC-MS · purified enzyme FMNH2 assay

Next step: plug in the wet-lab assay tables and prioritize decisive tests for our joint paper.

Workflow and tools by purpose

Question	Tools and evidence
Which read evidence enters assembly?	NanoPlot, Chopper, Filtlong
How are sequences reconstructed?	Autocycler/assemblers, dnaapler, Dorado
Where are genes and what are candidates?	Bakta, antiSMASH, DRAM, CANT-HYD, HADEG
How are genomes and proteins related?	TYGS, GTDB-Tk, ProteinOrtho/PoFF, IQ-TREE
How are local arrangements compared?	clinker and DNA Features Viewer

The TYGS genome phylogram



Blue labels: pseudo-bootstrap support (%) from 100 replicates.

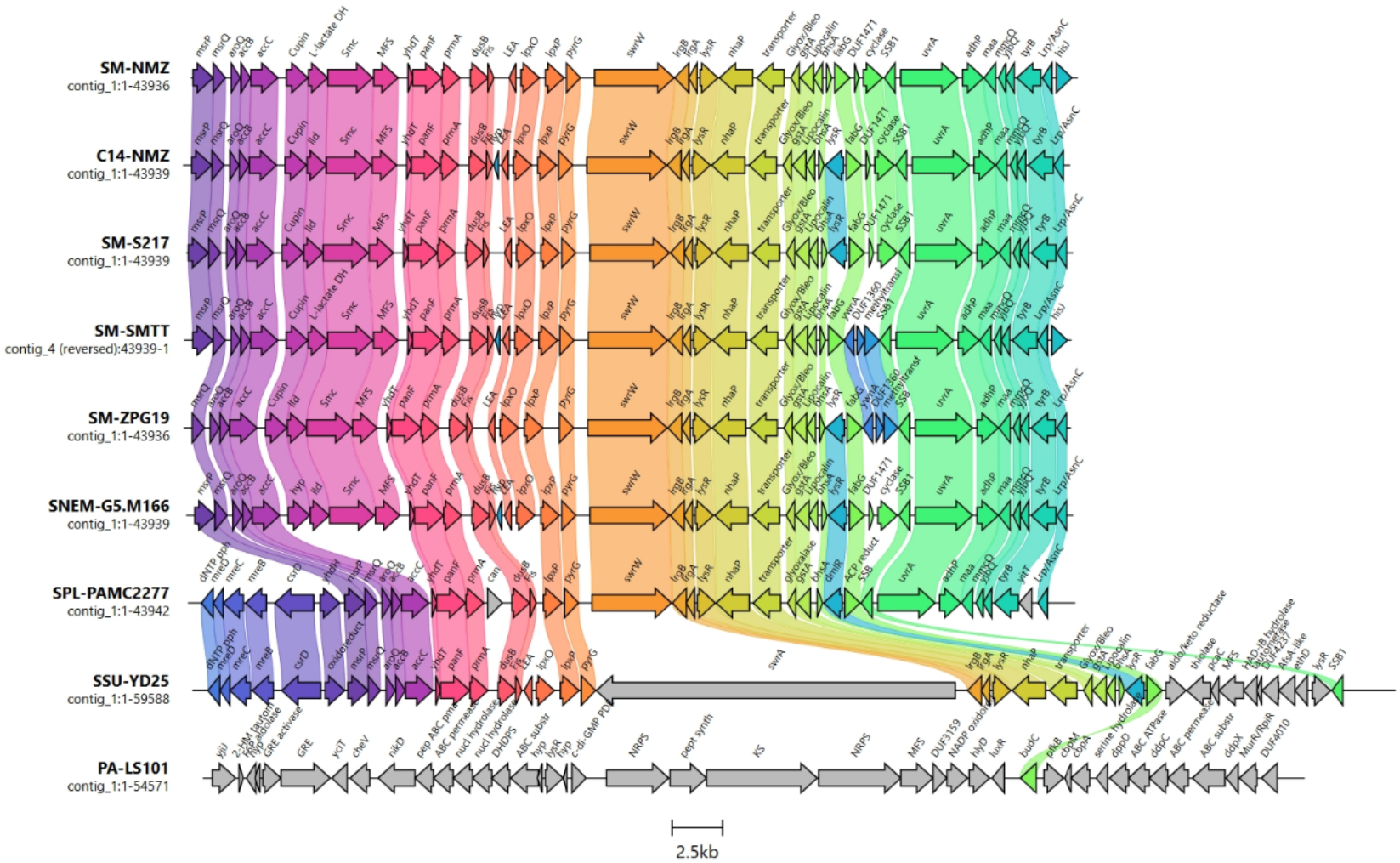
Taxonomic comparisons

dDDH(d4), %	SM-NMZ	C14-NMZ
S. marcescens ATCC 13880	90.7	73.7
S. nematodiphila DSM 21420	74.7	85.5
Direct focal comparison	74.2 (71.2–77.0)	Same comparison

GTDB ANI screening

SM 98.86% ANI (AF 0.947)
C14 98.22% ANI (AF 0.932)

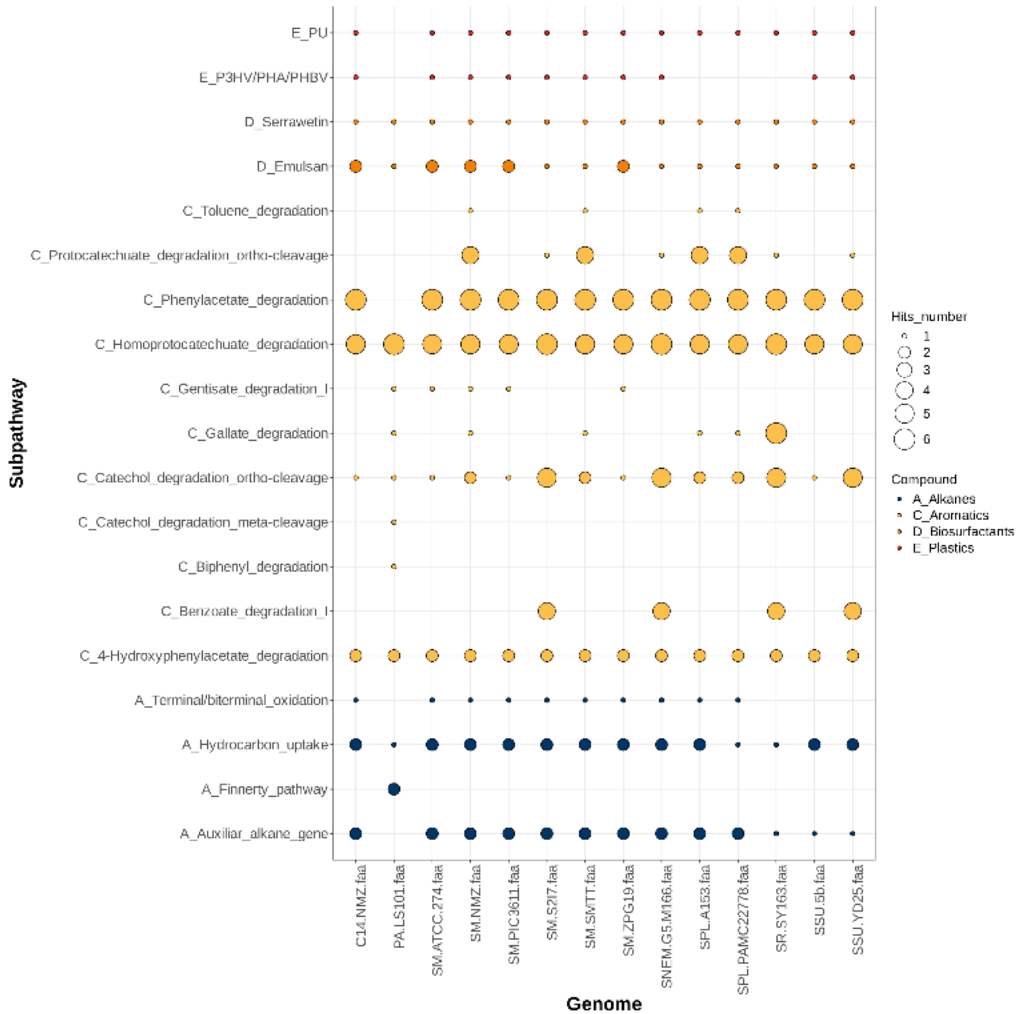
Reading the SwrW neighborhood



Arrows
Gene length and direction

Ribbons
Protein matches

Reading HADEG marker categories



Emulsan
Generic capsule markers

Toluene_degradation
Hydroxybenzoate-hydroxylase candidate

Bubble counts describe markers.
They do not measure pathway activity.

Chemical assignments and peak areas

Observation	What it can support
Exact mass and adduct	A candidate molecular composition
Fragmentation and reference match	Additional structural evidence
Authentic standard, when documented	A stronger identity comparison
Relative peak area	Detector signal under a stated denominator

Serrawettin W1 (= serratamolide A) and serratamolide C fit the SwrW candidate.

Additional SM aromatic candidates

C SM-NMZ hydroxybenzoate-hydroxylase / pcaK neighborhood

SM-NMZ | 2,461,766–2,467,540 bp | NNLMDO_011760–011775



D SM-NMZ protocatechuate-associated region

SM-NMZ | 3,121,817–3,130,237 bp | NNLMDO_014965–015005

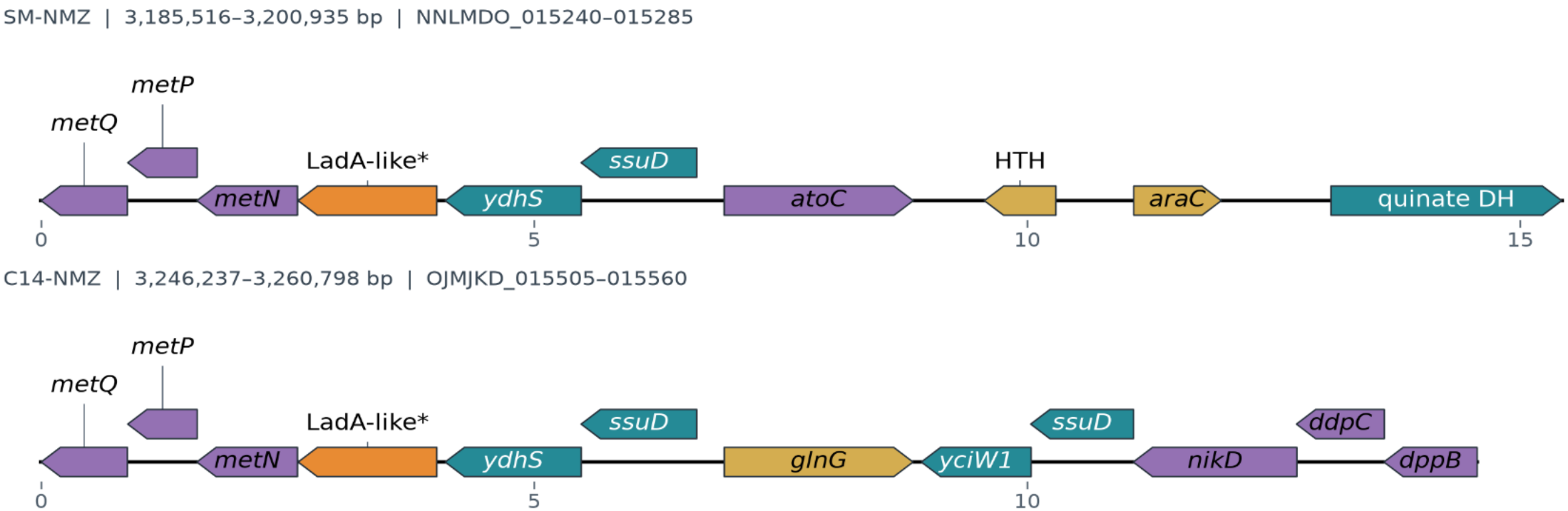


Scale in kb within each panel. Panels use different physical scales.

Additional candidates broaden the substrate hypotheses.

Some downstream steps remain unresolved.

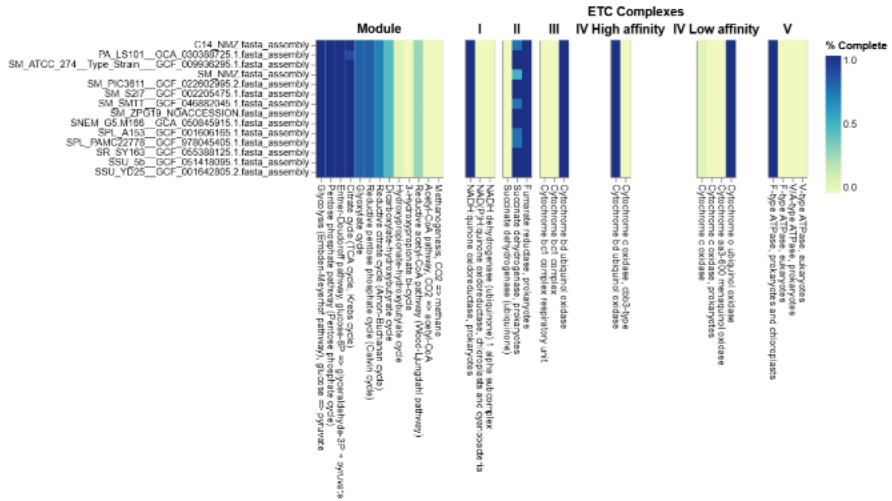
LadA-like genomic neighborhoods

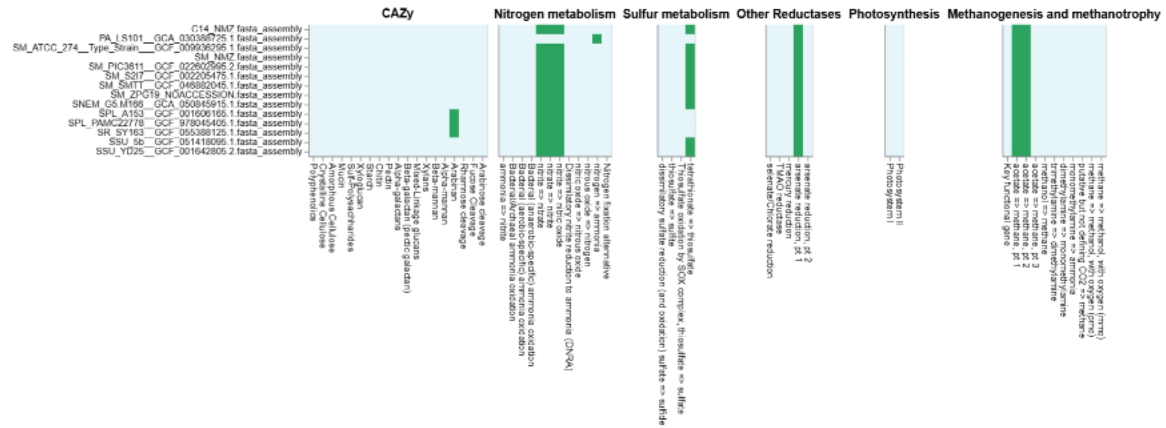


Gene arrows. Scale in kb from the first displayed CDS.

Source gene qualifier: ssuD. Candidate substrate remains unresolved.

Conserved metabolic background (DRAM)





DRAM Trait (only 3/98 vary within Serratia)	SM-NMZ	C14-NMZ	Variation Across 13 Serratia Genomes
Succinate dehydrogenase (ETC Complex II)	50%	75%	Ranges 50–100% module completeness across panel
CAZy GH43 arabinan cleavage call	Absent	Absent	Present only in S. marcescens KS10 & S. nematodiphila DSM 21420
Tetrathionate reductase marker (K08357)	Present	Present	Absent only in S. marcescens ATCC 13880 (present in 12/13)

95 of 98 DRAM traits (96.9%) are invariant across the 13 Serratia genomes; core metabolism is conserved.